

Data Analytics Capstone

Forecasting Short-Term Exchange Rate Movements and Volatility in Major Currency Pairs Using Macroeconomic Indicators and Machine Learning

Graded Assignment: Capstone Project Interim Report

Group 8

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QM640: Data Analytics Capstone

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Fall 2026 Term

August 22, 2026

GitHub Repository

All raw data, cleaning scripts, analysis notebooks, and figures referenced in this report are available at: <GitHub repository URL to be added>. The repository README documents the folder structure (raw/, processed/, notebooks/, figures/) and reproduction steps.

Introduction and Scope and Objectives

Background and Context

The foreign exchange (FX) market is the largest and most liquid financial market in the world, with global turnover exceeding USD 7 trillion per day (Bank for International Settlements, 2022). Currency values fluctuate continuously in response to macroeconomic fundamentals, central bank policy actions, and shifting investor sentiment. Meese and Rogoff (1983) demonstrated that structural macroeconomic models often fail to outperform a naive random-walk forecast over short horizons, motivating decades of research into alternative approaches, including machine learning (Bollerslev, 1986; Breiman, 2001; Chen & Guestrin, 2016; Hochreiter & Schmidhuber, 1997), sentiment analysis (Araci, 2019; Hutto & Gilbert, 2014; Tetlock, 2007), and central-bank event studies (Kuttner, 2001).

This project is directly relevant to corporate finance and treasury teams deciding when and how to hedge currency exposure, to FX traders and portfolio managers who need timely directional and volatility signals, and to policymakers monitoring exchange-rate stability.

Problem Statement and Purpose

Existing studies typically evaluate macroeconomic-fundamental models, machine-learning models, sentiment-based signals, and policy-event effects in isolation rather than within a single, comparable framework. This project addresses that gap: its purpose is to compare the effectiveness of macroeconomic-indicator-based models and machine-learning models in

predicting the short-term directional movement and volatility of USD/INR, EUR/USD, GBP/USD, and USD/JPY, and to evaluate whether financial-news sentiment and central-bank policy events materially improve forecast accuracy.

Research Questions

- **RQ1** - To what extent do macroeconomic indicators (interest-rate differential, inflation differential, trade balance, GDP growth rate) predict the short-term directional movement of the four currency pairs?
- **RQ2** - Do machine-learning models (Random Forest, XGBoost, LSTM) provide significantly better forecasting accuracy for exchange-rate direction and volatility than traditional econometric models (ARIMA, GARCH)?
- **RQ3** - Does incorporating financial-news sentiment significantly improve the predictive accuracy of exchange-rate direction models beyond macroeconomic indicators alone?
- **RQ4** - Is there a statistically significant relationship between central bank policy announcements and short-term spikes in exchange-rate volatility, and can the size of a policy “surprise” predict the magnitude of that spike?

Interim Project Status (Progress Snapshot)

Completed:

- Finalized problem statement, purpose, and four research questions (RQ1-RQ4), confirmed with the mentor.
- Completed literature review of 17 sources spanning FX forecasting, econometric volatility models, machine learning, NLP sentiment analysis, monetary policy event studies, and data quality/documentation practice.
- Identified and confirmed access to all public, non-Kaggle data sources (FRED, RBI DBIE, ECB SDW, Bank of England, IMF IFS, World Bank, GDELT Project).

- Collected daily historical exchange-rate series for all four currency pairs from FRED (2,143 clean daily observations per pair, Jan 2, 2018 - Jul 31, 2026) and the daily U.S. Federal Funds Effective Rate (DFF) as a first macroeconomic predictor.
- Completed initial data cleaning, exploratory data analysis, and minimum sample size computation for all four research questions.
- Built and evaluated a single-predictor pilot regression for RQ1 (see Preliminary Results).

In progress:

- Collecting the remaining RQ1 macroeconomic predictors (inflation differential, trade balance, GDP growth rate) to extend the pilot model into the full multi-predictor specification.
- Building ARIMA/GARCH baseline models and Random Forest/XGBoost/LSTM candidates for RQ2.
- Setting up the GDELT-based news sentiment extraction pipeline for RQ3.

Pending:

- Central bank policy-announcement calendar collection and the RQ4 event-study analysis.
- Full RQ2/RQ3 model training, hyperparameter tuning, and Diebold-Mariano/McNemar significance testing.
- Final cross-RQ synthesis, recommendations, and the practical dashboard-style deliverable.

Literature Survey

Literature Review Approach

Sources were identified using Google Scholar and the ACM Digital Library with search terms combining core concepts from each research question (e.g., “exchange rate forecasting” AND “machine learning”; “GARCH” AND “volatility”; “financial news” AND “sentiment analysis”; “monetary policy surprise” AND “volatility”; “data quality” AND “machine learning”; “statistical power” AND “sample size”), following the PRISMA-inspired process of defining the question, searching, and screening for relevance. Inclusion criteria favored foundational, highly-cited methodological papers directly underlying the statistical and machine-learning techniques proposed for RQ1-RQ4, as well as papers establishing best practice for data quality, sample-size justification, and dataset documentation, since this project's contribution is a comparative, well-documented application rather than a novel algorithm. Seventeen sources were selected; each is summarized in Table 1 below.

Summary of Key Literature (17 Sources)

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Finding	RQ Linkage
Meese & Rogoff (1983)	Int'l finance	Major exchange rates, 1970s-80s	Structural macro vs. random walk	Structural models do not outperform a random walk out-of-sample	RQ1, RQ2
Engle (1982)	Econometrics	U.K. inflation series	ARCH model	Introduced conditional heteroskedasticity modeling	RQ2
Bollerslev (1986)	Econometrics	General time series	GARCH model	Generalized ARCH to a parsimonious volatility model	RQ2
Breiman (2001)	Machine learning	General benchmark data	Random Forests	Ensemble of trees reduces variance, improves generalization	RQ1, RQ2
Hochreiter & Schmidhuber (1997)	Machine learning	Sequential/time-series data	LSTM networks	Gated recurrent architecture solves vanishing-gradient problem	RQ2
Chen & Guestrin (2016)	Machine learning	General benchmark data	XGBoost	Scalable gradient-boosted trees achieve state-of-the-art results	RQ2
Diebold & Mariano (1995)	Forecasting/stats	General forecast comparison	Diebold-Mariano test	Formal test for whether two forecasts differ significantly	RQ2

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Finding	RQ Linkage
Hutto & Gilbert (2014)	NLP	Social media text	VADER sentiment	Rule-based lexicon achieves strong accuracy on informal text	RQ3
Araci (2019)	NLP/Finance	Financial text corpora	FinBERT	Domain-adapted transformer improves financial sentiment classification	RQ3
Tetlock (2007)	Behavioral finance	U.S. news/stock market	Media pessimism index vs. returns	Media sentiment predicts short-term downward market pressure	RQ3
Kuttner (2001)	Monetary economics	Fed funds futures market	Event-study regression	Unexpected policy changes move rates far more than expected ones	RQ4
Rahm & Do (2000)	Data engineering	General data warehousing	Taxonomy of data-quality problems	Classifies single/multi-source and schema/instance-level data errors	Data Cleaning
O'Brien, Sukumar & Helfert (2013)	IT/data governance	Case studies, 2 firms	Data-quality cost taxonomy (People-Process-Data)	Poor data quality has direct, measurable operational and financial cost	Data Description
Button et al. (2013)	Research methodology	49 neuroscience meta-analyses	Power/PPV analysis	Low statistical power inflates effect sizes and reduces reliability of findings	Sample Size (all RQs)
Geburu et al. (2018)	ML data documentation	General ML datasets	Datasheets for Datasets schema	Standardized dataset documentation improves transparency and reuse	Data Description
Bender & Friedman (2018)	NLP data documentation	Language datasets	Data Statements schema	Documenting speaker/annotator demographics mitigates NLP system bias	RQ3 (sentiment data)
Torralba & Efros (2011)	Computer vision	12 image-recognition datasets	"Name That Dataset" classifier	Benchmark datasets carry a detectable "signature" bias limiting generalization	Data Description / Limitations

Table 1. Literature relevance matrix.

Foundations of the Forecasting Puzzle

Meese and Rogoff (1983) remains the foundational, cautionary reference for this project, framing RQ1 and RQ2 as a comparative test rather than an assumption that any single model will trivially succeed. The preliminary pilot result reported below is consistent with this literature.

Volatility Modeling, Machine Learning, and Sentiment

Engle (1982) and Bollerslev (1986) provide the econometric baseline against which RQ2's machine-learning models (Breiman, 2001; Hochreiter & Schmidhuber, 1997; Chen & Guestrin, 2016) are benchmarked, with Diebold and Mariano (1995) supplying the significance test. Hutto and Gilbert (2014) and Araci (2019) provide two complementary sentiment-scoring approaches for RQ3, motivated by Tetlock's (2007) behavioral-finance evidence that sentiment carries incremental predictive information.

Monetary Policy and Event Studies

Kuttner (2001) directly motivates the RQ4 event-study design and its “policy surprise” construct, providing evidence that surprises, not expected changes, move markets.

Data Quality, Documentation, Bias, and Statistical Power

A second cluster of literature underlies the Analysis and Data Description sections of this report rather than the forecasting models themselves. Rahm and Do (2000) provide a classification of data-quality problems (single- vs. multi-source, schema- vs. instance-level) that directly frames the Data Cleaning subsection below. O'Brien, Sukumar, and Helfert (2013) show, through case studies of two large organizations, that poor data quality carries direct and indirect operational costs, underscoring why this project documents its cleaning decisions explicitly rather than treating them as an afterthought. Gebru et al. (2018) and Bender and Friedman (2018) provide standardized documentation schemas - “datasheets for datasets” and “data statements,” respectively - that this report draws on when describing the provenance, composition, and collection process of the FX/macro data (Gebru et al.) and the planned GDELT news-sentiment text data (Bender & Friedman), since the latter is linguistic data with its own population and bias considerations. Torralba and Efros (2011) demonstrate empirically that benchmark datasets carry a detectable “signature” that limits cross-dataset generalization; this motivates the limitation, noted below, that models trained on 2018-2026 data may not

generalize to structurally different monetary regimes. Finally, Button et al. (2013) show across 49 neuroscience meta-analyses that underpowered studies inflate effect sizes and reduce the reliability of findings (the “winner's curse”) - the statistical justification underlying the minimum sample size computation performed for each research question in the Analysis section.

Data Description

Data Source(s) and Access

- Federal Reserve Economic Data (FRED) - daily exchange rates and U.S. interest rates: fred.stlouisfed.org (Federal Reserve Bank of St. Louis, n.d.)
- RBI Database on Indian Economy (DBIE) - India policy rate, inflation: dbie.rbi.org.in
- European Central Bank Statistical Data Warehouse (SDW) - Eurozone rates: sdw.ecb.europa.eu
- Bank of England Statistical Interactive Database - UK rates: bankofengland.co.uk/statistics
- IMF International Financial Statistics (IFS) - trade balance and GDP data: data.imf.org
- World Bank Open Data - GDP growth rates: data.worldbank.org
- GDELT Project - public news tone/sentiment data (non-Kaggle): gdeltproject.org

Confirmed Series for Remaining RQ1 Predictors

Since the pilot regression, each of the eight sources above was probed directly for the specific series needed to complete the RQ1 interest-rate and inflation differentials. FRED mirrors most of this international data directly, which simplifies collection considerably. Table 2a lists the confirmed series and their live coverage as of this checkpoint.

Predictor	Country/Region	FRED Series ID	Coverage Confirmed
Short-term policy/interbank rate	United States	DFF	Daily, live through Aug 2026 (already in use)
Short-term interbank rate	India	IRSTCI01INM156N	Monthly, live through Jun 2026
Short-term interbank rate	Euro area	IRSTCI01EZM156N	Monthly, live through Jan 2026
Short-term interbank rate (SONIA)	United Kingdom	IUDSOIA	Daily, live through Jul 2026
Short-term interbank rate	Japan	IRSTCI01JPM156N	Monthly, live through Jun 2026
CPI (for inflation differential)	United States	CPIAUCSL	Monthly, live through Jul 2026
CPI (for inflation differential)	India	INDCPIALLMINMEI	Monthly, but stops Mar 2025 (stale)
HICP (for inflation differential)	Euro area	CP0000EZ19M086NEST	Monthly, live through Jul 2026
CPI (for inflation differential)	United Kingdom	GBRCPIALLMINMEI	Monthly, but stops Mar 2025 (stale)
CPI (for inflation differential)	Japan	JPNCPIALLMINMEI (and all FRED-mirrored alternatives tested)	Discontinued - last observation Jun 2021
Trade balance	United States	BOPGSTB	Monthly, live through Jun 2026

Table 2a. Confirmed data series for the remaining RQ1 macroeconomic predictors.

Two data-quality findings emerged from this check, consistent with the data-quality literature reviewed above (Rahm & Do, 2000; O'Brien et al., 2013): first, FRED's mirrored India and UK CPI series are roughly a year and a half stale (last updated March 2025), so the inflation differential for USD/INR and GBP/USD will need a supplementary source (RBI DBIE and UK ONS, respectively) to stay current; second, every FRED-mirrored Japan CPI series tested stopped updating in June 2021, so the USD/JPY inflation differential will require pulling CPI directly from Japan's Statistics Bureau (stat.go.jp) or the OECD Data Explorer rather than FRED. Trade balance and GDP growth series for India, the Euro area, the UK, and Japan were not yet located on FRED at monthly/quarterly frequency and remain to be sourced from IMF IFS or World Bank, consistent with the original plan.

Dataset Overview

As of this checkpoint, the exchange-rate leg of the dataset has been fully collected and cleaned for all four currency pairs, summarized in Table 2. Each series was pulled from

FRED's public CSV endpoint for January 1, 2018 through August 1, 2026 (2,239 calendar-day requests per series); after removing non-trading days (96 rows per series), 2,143 clean daily observations remain per pair (8,572 pooled). The daily U.S. Federal Funds Effective Rate (DFF) was collected over the same window (3,135 clean observations).

Field	Value
Number of records (rows)	2,143 per currency pair (8,572 pooled); 3,135 for DFF
Number of variables (columns)	4 variables collected to date (Table 3); 8 further variables planned (interest_rate_diff, inflation_diff, trade_balance, gdp_growth_rate, news_sentiment_score, cb_announcement_flag, policy_surprise, currency_pair)
Time period	January 2, 2018 - July 31, 2026
Unit of analysis	Currency pair x trading day
Target variable(s)	Daily log return (RQ1, RQ2) and realized volatility (RQ2, RQ4)

Table 2. Dataset overview.

Data Dictionary

Variable Name	Definition	Datatype	Range	Missing-Value Handling	Notes
exchange_rate	Official daily closing level, local currency per USD	Float	USD/INR 63.38-96.82; EUR/USD 0.96-1.25; GBP/USD 1.07-1.43; USD/JPY 102.52-163.86	Rows with no published rate dropped	Source: FRED DEXINUS, DEXUSEU, DEXUSUK, DEXJPUS
log_return	$\log(\text{rate}_t) - \log(\text{rate}_{t-1})$	Float	Approx. -2% to +2% (Figure 2)	First observation per series dropped	Derived
dff	U.S. Federal Funds Effective Rate	Float	0.05 - 5.33 (2018-2026)	None observed	Source: FRED DFF
dff_chg_lag1	1-day lagged change in DFF	Float	-0.85 to +0.75 percentage points; 0 on most days	First 2 obs. per series dropped	Derived; used in pilot regression

Table 3. Data dictionary (variables collected to date).

Dataset Documentation (Datasheet-Style Summary)

Following the “datasheets for datasets” schema (Geburu et al., 2018), the FX and macro data are summarized below to support transparent reuse:

- **Motivation:** Collected specifically to test RQ1 (macroeconomic drivers of currency movement); no gap-filling of a pre-existing internal dataset was involved.
- **Composition:** Each instance is one currency pair on one trading day; the dataset is the full population of published FRED observations in the window, not a sub-sample, so no sampling bias was introduced at this stage.
- **Collection process:** Data was pulled programmatically via FRED's public CSV API (not manually curated), directly observable (official closing rates), with no human subjects involved.
- **Preprocessing:** Non-numeric/missing rows were dropped and log returns derived; the raw downloaded CSVs are retained alongside the processed files in the GitHub repository.
- **Known limitation:** Consistent with Torralba and Efros (2011), a model trained only on the 2018-2026 regime may not generalize to structurally different monetary conditions (e.g., a return to near-zero global interest rates); this is logged as a limitation below.

The planned GDELT news-sentiment text data (for RQ3) will additionally require a *data statement* in the sense of Bender and Friedman (2018), since it is linguistic data with its own speaker population: at minimum, the language variety (English-language news), curation rationale (headlines mentioning the four currency pairs/countries), and time period will be documented once collected, so that any skew in which news outlets or regions are represented is visible to downstream users of the sentiment feature.

GitHub Data Availability Statement

The raw and processed datasets are available in the GitHub repository under: Raw data: /data/raw/; Processed/clean data: /data/processed/; Code/notebooks: /notebooks/. The pilot data-collection and regression script is reproduced in Appendix A.

Screenshots / Evidence

Figure 1 previews the collected exchange-rate data, plotted as an indexed level (Jan 2, 2018 = 100) so that all four pairs are visually comparable despite their different price scales.

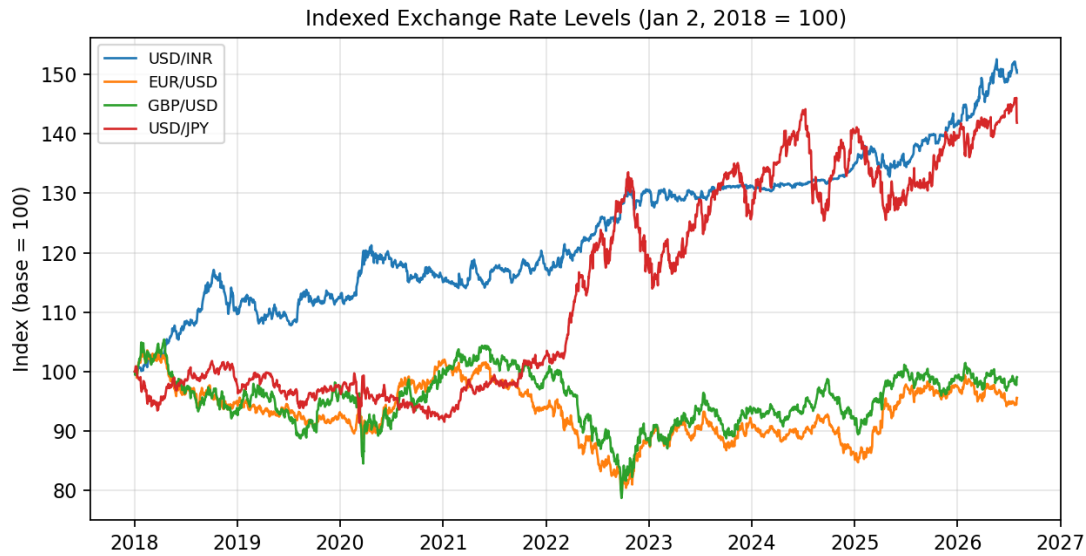


Figure 1. Indexed daily closing levels for all four currency pairs, 2018-2026 (source: FRED).

Analysis

Data Cleaning

Each raw FRED series was checked for missing values, duplicate dates, and outliers before computing log returns. Using Rahm and Do's (2000) classification of data-quality problems, the missingness observed here (96 of 2,239 rows per series, 4.3%) is a single-source, instance-level “missing values” problem, and is structural (non-trading days) rather than random, so it was resolved by row deletion rather than imputation. No duplicate dates were found (Rahm and Do's “uniqueness violation” check), consistent with FRED series being pre-deduplicated by construction. Extreme daily moves (beyond 3 standard deviations) were flagged but retained, since for RQ2 and RQ4 these tail events are analytically important rather than data errors. Table 4 logs each issue and its treatment.

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Missing values	exchange_rate (all 4 pairs)	Null check after CSV parse	Row deletion (96/2,239 rows, 4.3%)	Missingness is structural (non-trading days), not random
Outliers / tail events	log_return (all 4 pairs)	3-sigma threshold on daily returns	Flagged, not removed	Tail events are the volatility spikes RQ2/RQ4 are designed to study
Duplicates	date (all series)	Duplicate-key / uniqueness check	None found	FRED daily series are pre-deduplicated by construction

Table 4. Data cleaning log.

EDA Results

Pair	N	Ann. Vol. (%)	Return Skew	Return Kurtosis
USD/INR	2,143	5.04	0.26	4.29
EUR/USD	2,143	7.04	0.10	1.69
GBP/USD	2,143	8.62	0.02	3.04
USD/JPY	2,143	8.90	-0.64	5.12

Table 5. Descriptive statistics of daily log returns by currency pair, 2018-2026.

As shown in Table 5, annualized volatility ranges from 5.04% (USD/INR) to 8.90% (USD/JPY). USD/INR's comparatively low volatility is consistent with the Reserve Bank of India's managed-float intervention practice, while GBP/USD and USD/JPY show the highest volatility. All four return series show excess kurtosis (1.7 to 5.1, versus 0 for a normal distribution), meaning large moves occur far more often than a normal distribution would predict, as shown in Figure 2. This fat-tailed, volatility-clustering behavior is the empirical motivation for including GARCH as a baseline in RQ2, rather than a constant-variance model.

Minimum Sample Size Computation per Research Question

Following Button et al. (2013), who show that underpowered studies inflate effect-size estimates and reduce the reliability of findings, each research question's minimum sample size is computed at the standard 95% confidence / 80% power convention before any modeling is finalized, using the method appropriate to its analytic design (Table 6).

RQ	Method	Key Parameters	Min. N	Unit
RQ1	Multiple Regression (Green's Rule)	$k = 5$ predictors (interest-rate diff., inflation diff., trade balance, GDP growth, lagged return); $n \geq 50 + 8k$	$n \geq 90$	Monthly, per pair
RQ2	Two-proportion power test	$\alpha = 0.05$, Power = 0.80, p_1 (econometric) = 0.55, p_2 (ML) = 0.65	$n \geq 373$ /grp	Daily forecast days
RQ3	Two-proportion power test	$\alpha = 0.05$, Power = 0.80, $p_1 = 0.65$, $p_2 = 0.72$	$n \geq 687$ /grp	Daily forecast days
RQ4	Correlation-based power formula	$\alpha = 0.05$, Power = 0.80, expected $r = 0.30$; $n = [2.8 / (0.5 \ln((1+r)/(1-r)))]^2 + 3$	$n \geq 85$ events	Per CB policy event

Table 6. Minimum sample size per research question.

The two-proportion calculations (RQ2, RQ3) follow the classical result that, for 95% confidence and 80% power, the hypothesized effect must be at least 2.8 standard errors from the null comparison point (Gelman & Hill, 2007, Ch. 20). The final sample size for the project is the maximum across RQ1-RQ4, $n \geq 687$ daily observations (driven by RQ3). The 2,143 observations already collected per currency pair comfortably exceed this threshold, and 8 years of history further ensures the RQ4 event study has access to well over 85 central-bank announcement dates (four central banks $\times$ ~ 8 meetings/year $\times$ ~ 8 years is approximately 256 events).

Visualizations and Respective Insights

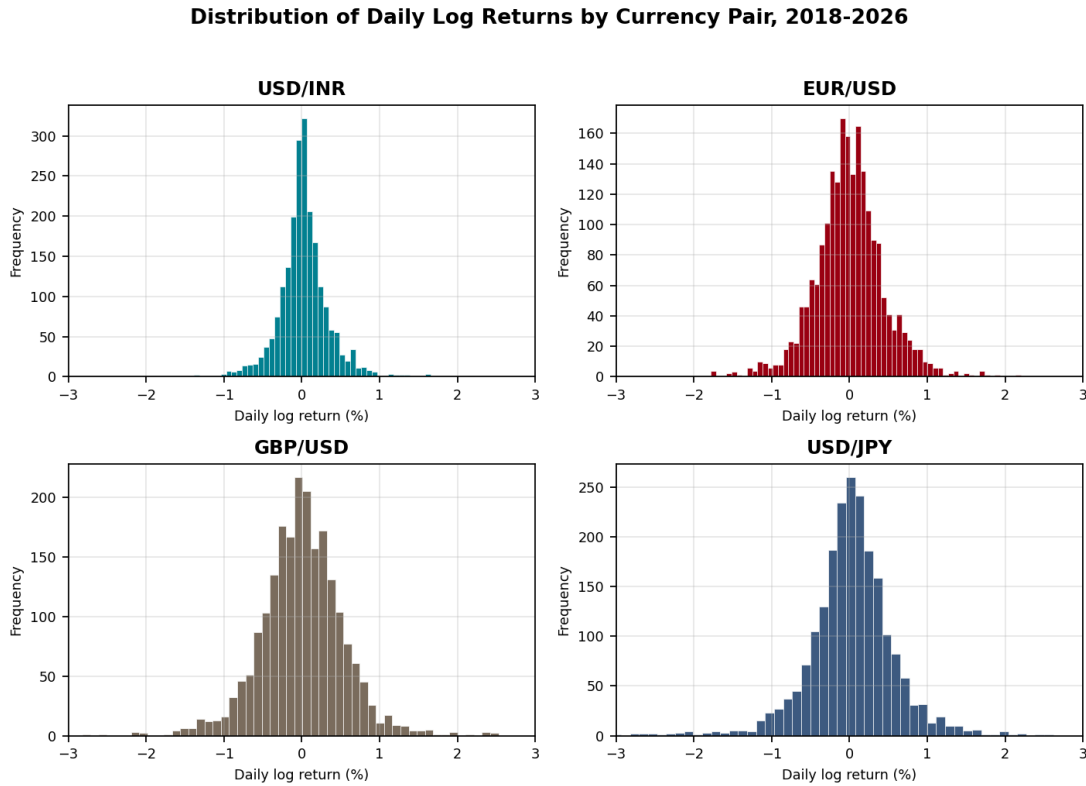


Figure 2. Distribution of daily log returns for all four currency pairs, 2018-2026 - note the fat tails relative to a normal distribution in every pair.

Figure 2 shows a sharp central peak with visibly fat tails relative to a normal curve in all four currency pairs, confirming the excess-kurtosis finding in Table 5 (kurtosis 1.7 to 5.1 across pairs) and directly motivating the GARCH baseline used for RQ2. USD/JPY shows the most pronounced tail asymmetry (skew = -0.64), consistent with occasional sharp yen appreciation events (e.g., Bank of Japan policy shifts) outweighing the more numerous small depreciation days.

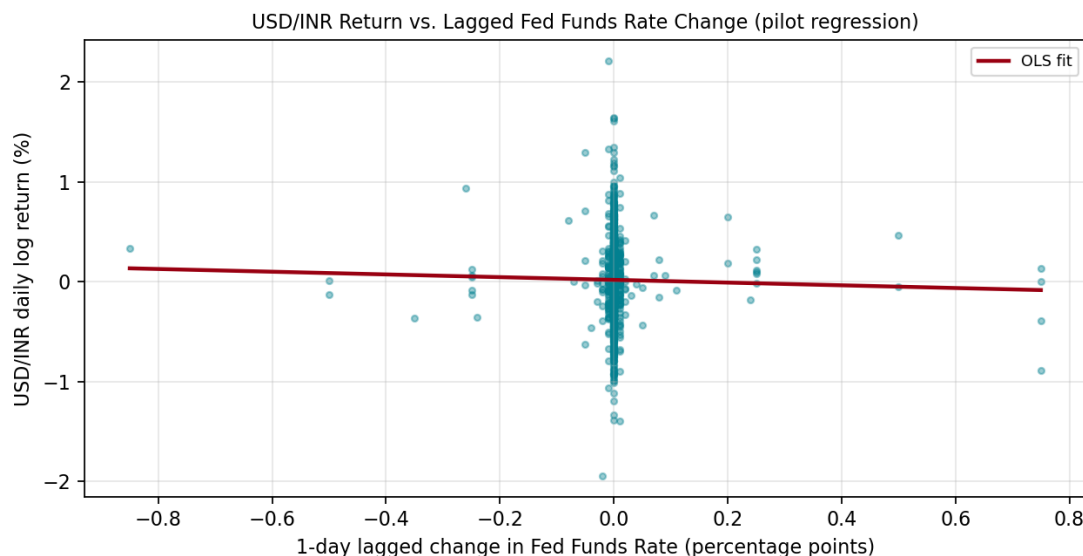


Figure 3. USD/INR daily return vs. 1-day lagged change in the U.S. Federal Funds Rate, with OLS fit.

Figure 3 visualizes the RQ1 pilot regression reported below: the fitted line is nearly flat and the point cloud is highly dispersed at every value of the lagged rate change, visually confirming the near-zero R-squared and non-significant coefficient. This is the key insight motivating the shift to a full multi-predictor model rather than a single U.S.-only policy variable.

Figure/Table	What it Shows	Key Insight	Why it Matters	Next Step
Figure 1	Indexed levels, 4 pairs	INR and JPY depreciated most vs. USD since 2018; EUR/GBP roughly flat	Confirms meaningful cross-pair variation for RQ1 modeling	Use as context for RQ1 coefficients
Figure 2 / Table 5	Return distribution and cross-pair volatility, all 4 pairs	Fat tails and excess kurtosis in all 4 pairs; USD/JPY most left-skewed	Justifies GARCH baseline in RQ2	Fit per-pair GARCH(1,1) and compare conditional volatility to realized volatility
Figure 3	Return vs. lagged Fed-rate change	Flat fit, high dispersion, no visible relationship	Confirms the pilot's null result is not an artifact of a hidden nonlinearity	Add remaining 3 macro predictors and re-test

Table 7. Visualization insight summary.

Modelling

Choice of Models with Justification

- RQ1 - Multiple linear regression (OLS) with a Random Forest regression robustness check: chosen because the RQ1 hypothesis concerns the linear relationship implied by interest-rate and purchasing-power parity theory, with Random Forest testing for nonlinearity (Meese & Rogoff, 1983; Breiman, 2001). VIF checks will screen for multicollinearity among the four macro predictors.
- RQ2 - ARIMA/GARCH (Engle, 1982; Bollerslev, 1986) vs. Random Forest, XGBoost, and LSTM (Breiman, 2001; Chen & Guestrin, 2016; Hochreiter & Schmidhuber, 1997), compared via the Diebold-Mariano test (Diebold & Mariano, 1995) and time-series cross-validation (no random shuffling, to avoid look-ahead leakage).
- RQ3 - VADER and FinBERT sentiment scoring (Hutto & Gilbert, 2014; Araci, 2019) added as a feature to the best RQ2 model, compared using McNemar's test, consistent with Tetlock's (2007) rationale for sentiment's incremental value.
- RQ4 - Event-study paired-difference testing plus logistic regression on policy surprise (Kuttner, 2001), mirroring the established event-study design for monetary-policy shocks.

Features Included and Feature Engineering

Feature	Original/Engineered	Type	Reason for Inclusion	Used in Model(s)
log_return	Engineered (from exchange_rate)	Continuous	Standard, stationarity-friendly FX outcome measure	RQ1, RQ2, RQ3
realized_volatility	Engineered (rolling std. dev.)	Continuous	Captures volatility clustering confirmed in EDA	RQ2, RQ4
interest_rate_diff	Engineered (domestic - U.S. rate)	Continuous	Directly implied by interest-rate parity theory	RQ1
dff_chg_lag1	Engineered (1-day lagged DFF change)	Continuous	Pilot proxy for policy-rate movement	RQ1 (pilot)
news_sentiment_score	Engineered (VADER/FinBERT on GDELT)	Continuous	Tests the RQ3 sentiment hypothesis	RQ3
cb_announcement_flag	Engineered (policy calendars)	Binary	Identifies event-study windows for RQ4	RQ4

Table 8. Feature set and feature engineering.

Preliminary Results

RQ1 (pilot).

A single-predictor OLS regression was run of USD/INR daily log return on the 1-day lagged change in the U.S. Federal Funds Effective Rate ($n = 2,142$ daily observations, exceeding the $n \geq 90$ minimum from Table 6). The pilot model explains essentially none of the variation in daily returns ($R\text{-squared} = 0.00044$, adjusted $R\text{-squared} < 0$), with a coefficient of -0.0014 ($SE = 0.0014$, $p = 0.331$), visualized in Figure 3. This null result is consistent with Meese and Rogoff (1983) and confirms the decision to specify RQ1 as a multi-predictor model rather than a single country's raw policy rate.

RQ2-RQ4 (pending).

Baseline ARIMA/GARCH fitting and ML model training for RQ2 are in progress but not yet complete; RQ3 (sentiment) and RQ4 (event study) remain in the data-collection phase. No preliminary accuracy figures are reported for these to avoid presenting partial, potentially misleading comparisons.

Preliminary Model Performance

Table 9 reports the one model fit to date (the RQ1 pilot regression); the RQ2 rows are included to show the planned comparison structure and will be populated as those models are completed.

Model	Feature Set	Validation Approach	R-squared	p-value (key coef.)	Key Takeaway
OLS (pilot)	dff_chg_lag1 only	In-sample, single predictor	0.00044	0.331	No significant single-predictor effect; motivates full multi-predictor RQ1 model

Model	Feature Set	Validation Approach	R-squared	p-value (key coef.)	Key Takeaway
ARIMA/GARCH baseline (RQ2)	Pending	Rolling window (planned)	Pending	Pending	Data collection/training in progress
Random Forest / XGBoost / LSTM (RQ2)	Pending	Time-series CV (planned)	Pending	Pending	Not yet started

Table 9. Preliminary model performance comparison.

Interim Limitations and Risks

- The RQ1 pilot uses only one macroeconomic predictor and U.S.-side data only; it is not yet a test of the full RQ1 hypothesis, and its null result should not be over-interpreted until the multi-predictor model is complete.
- Inflation, trade-balance, and GDP data are lower-frequency (monthly/quarterly) than the daily FX series, and aligning frequencies without introducing look-ahead bias will require careful lagging, planned for Week 4-5.
- GDELT-based sentiment and central-bank policy-calendar data have not yet been collected; schedule risk exists if API access or data volume proves larger than expected, mitigated by starting this collection immediately (Week 3 of the project's 10-week phased plan).
- The current pilot does not yet implement train/validation/test splitting or cross-validation; this will be added once the full RQ1 feature set is assembled.
- As noted in the Literature Survey (Torralba & Efros, 2011), a model trained only on the 2018-2026 regime may carry a dataset-specific “signature” that limits generalization to structurally different monetary conditions.

Next Steps (for Final Report)

- Collect inflation differential, trade balance, and GDP growth data and extend the RQ1 model to the full multi-predictor specification with VIF checks.

- Fit ARIMA/GARCH baselines (AIC/BIC order selection) and Random Forest/XGBoost/LSTM models for RQ2; run the Diebold-Mariano test for model comparison.
- Build the GDELT sentiment pipeline (VADER and FinBERT) and evaluate the RQ3 sentiment-augmented model against the best RQ2 model using McNemar's test.
- Collect central bank policy-announcement calendars and conduct the RQ4 event-study analysis.
- Synthesize findings across RQ1-RQ4, finalize the GitHub repository, and prepare recommendations and the final presentation.

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Appendix

Appendix A: Data Collection and Pilot Regression Code

The following Python script was used to download the FRED series, compute daily log returns, run the RQ1 pilot regression reported above, and generate Figures 1-3. It is also included in the GitHub repository under /notebooks/.

```
import pandas as pd, numpy as np, statsmodels.api as sm
pairs = {"USD/INR": "DEXINUS.csv", "EUR/USD": "DEXUSEU.csv",
         "GBP/USD": "DEXUSUK.csv", "USD/JPY": "DEXJPUS.csv"}
series = {}
for name, f in pairs.items():
    df = pd.read_csv(f, parse_dates=["observation_date"])
    col = df.columns[1]
    df[col] = pd.to_numeric(df[col], errors="coerce")
    df = df.dropna(subset=[col]).reset_index(drop=True)
    df["log_ret"] = np.log(df[col]).diff()
    series[name] = df

dff = pd.read_csv("DFF.csv", parse_dates=["observation_date"])
dff.columns = ["date", "dff"]
dff["dff"] = pd.to_numeric(dff["dff"], errors="coerce")
dff = dff.dropna(); dff["dff_chg"] = dff["dff"].diff()

inr = series["USD/INR"][["observation_date", "log_ret"]].rename(
    columns={"observation_date": "date"})
merged = pd.merge(inr, dff[["date", "dff_chg"]], on="date", how="inner")
merged["dff_chg_lag1"] = merged["dff_chg"].shift(1)
merged = merged.dropna()

X = sm.add_constant(merged["dff_chg_lag1"])
model = sm.OLS(merged["log_ret"], X).fit()
print(model.summary())
```